

Optiver Cognitive Assessment - Section 1

Instructions: Choose the best answer. No calculators. Work quickly but carefully.

Question 1. A market-making bot sees three independent signals S1, S2, S3. Exactly two of them are reliable today, but you do not know which two. A reliable signal predicts the next tick correctly with probability 0.72; an unreliable one with probability 0.48. All three currently say UP. Before seeing them, each pair of reliable signals was equally likely. What is the probability the next tick is UP?

- A. 0.600
- B. 0.640
- C. 0.672
- D. 0.704
- E. 0.720

Question 2. A trader must compute a fair adjustment quickly. Start with 2400. Increase by 12.5%, then decrease the result by 20%, then add back 8% of the original 2400. What is the final value?

- A. 2352
- B. 2376
- C. 2400
- D. 2472
- E. 2496

Question 3. Five orders A, B, C, D, E are processed one at a time. A is before C. D is immediately after B. E is not first or last. C is not adjacent to D. Which order is possible?

- A. B D E A C
- B. E A C B D
- C. A B D E C
- D. B D A C E
- E. A E B D C

Question 4. A robot has counters (x,y) starting at (0,0). In one move it may do exactly one of: add 3 to x, add 4 to y, or transfer 2 from x to y if $x \geq 2$. It must end at (7,10). What is the minimum number of moves?

- A. 5
- B. 6
- C. 7
- D. 8
- E. Impossible

Question 5. In the string below, count tokens that contain exactly two letters and one digit, in any order, with no repeated character. Ignore tokens containing punctuation. AX7 4BB C9D Q22 R5T Z0Y AB-3 M1N 77A K2K P8L

- A. 4
- B. 5

- C. 6
- D. 7
- E. 8

Question 6. Sequence: 3, 8, 18, 35, 61, 98, ?. The increments are generated by alternating +5, +10, +17, +26, +37, ... where the second differences of increments rise by 2 each time. What comes next?

- A. 142
- B. 146
- C. 148
- D. 150
- E. 154

Question 7. A shuffled deck has 6 red and 4 blue cards. You draw 3 without replacement. You win if the third card is red and at least one of the first two cards is blue. What is the win probability?

- A. $1/5$
- B. $4/15$
- C. $3/10$
- D. $1/3$
- E. $2/5$

Question 8. Four analysts each make one statement about a hidden integer n from 1 to 20. P: n is even. Q: n is divisible by 3. R: n is greater than 10. S: n is prime. Exactly two statements are true. How many possible values can n have?

- A. 5
- B. 6
- C. 7
- D. 8
- E. 9

Question 9. A quote moves from 125 to 119, then recovers to 123.76. Compared with the original 125, what is the net percentage change?

- A. -1.20%
- B. -0.992%
- C. -0.80%
- D. +0.64%
- E. +3.76%

Question 10. Three stacks contain chips: $A=7$, $B=0$, $C=0$. A move transfers exactly 2 chips from one stack to another, or exactly 3 chips from one stack to another. No stack may become negative. Can you reach $A=1$, $B=3$, $C=3$, and what is the minimum number of moves?

- A. 2 moves
- B. 3 moves

C. 4 moves

D. 5 moves

E. Impossible

Optiver Cognitive Assessment - Section 2

Instructions: Choose the best answer. No calculators. Work quickly but carefully.

Question 1. A feed lists codes. Count only codes whose first and last characters are letters, whose middle two characters form a two-digit number divisible by 4, and whose letters are different. A12B C08C D16F 9H24J K20L M36N P40P Q44R S52T U6V W88X

- A. 5
- B. 6
- C. 7
- D. 8
- E. 9

Question 2. A box has 5 fair coins and 1 double-headed coin. One coin is chosen uniformly and flipped 4 times. All 4 flips are heads. What is the probability the chosen coin is double-headed?

- A. $8/13$
- B. $16/21$
- C. $16/31$
- D. $1/2$
- E. $21/26$

Question 3. A position gains 30%, loses 25%, gains 20%, then loses 10%. Starting from 10,000, what is the final value?

- A. 10,350
- B. 10,530
- C. 10,800
- D. 11,000
- E. 11,700

Question 4. Six people sit in a row. L is left of M but not necessarily adjacent. N is exactly two seats right of L. O is not adjacent to M. P is at one end. Q is immediately left of O. Which person must be in seat 4 if P is in seat 6? Seats are numbered left to right.

- A. M
- B. N
- C. O
- D. Q
- E. Cannot be determined

Question 5. Find the missing term: 2, 6, 15, 31, 56, 92, ?. The successive differences are 4, 9, 16, 25, 36, then the next square with the same rule.

- A. 135
- B. 141

- C. 145
- D. 149
- E. 156

Question 6. A task has states 0 through 9. From state k you may move to $k+2$, $k+3$, or $k*2$, but cannot exceed 9. Starting at 1, what is the minimum number of moves to visit both 7 and 9 at least once?

- A. 2
- B. 3
- C. 4
- D. 5
- E. Impossible

Question 7. Two dice are rolled. You are told the product is divisible by 6. What is the probability that the sum is at least 8?

- A. $5/12$
- B. $6/17$
- C. $7/17$
- D. $8/17$
- E. $1/2$

Question 8. Cards A, B, C, D each have a hidden integer. The four integers are 2, 3, 5, 8 in some order. $A+B$ is odd. $B+C$ is prime. $C+D$ is odd. A is not 8. What is B?

- A. 2
- B. 3
- C. 5
- D. 8
- E. Cannot be determined

Question 9. A price is quoted as 1.6% above fair value. A correction drops the quote by 1.25% of the quoted price. Relative to fair value, the corrected quote is approximately:

- A. 0.33% high
- B. 0.35% high
- C. 0.37% high
- D. 0.39% high
- E. 0.41% high

Question 10. In the list 18, 27, 42, 54, 63, 72, 81, 96, 108, 126, count numbers divisible by 9 but not by 6, plus numbers divisible by 6 but not by 9. What is the count?

- A. 5
- B. 6
- C. 7
- D. 8

Optiver Cognitive Assessment - Section 3

Instructions: Choose the best answer. No calculators. Work quickly but carefully.

Question 1. A signal is green on 60% of days. If the signal is green, a model is correct with probability 0.75; if red, it is correct with probability 0.55. The model was correct yesterday. What is the probability the signal was green?

- A. 15/26
- B. 9/13
- C. 45/67
- D. 3/5
- E. 5/8

Question 2. There are three boxes labeled X, Y, Z. One label is correct, two are incorrect. The boxes contain: only red, only blue, and mixed red/blue. Label X says red, label Y says blue, label Z says mixed. You draw one red ball and then one blue ball from box X without replacement. What must box X contain?

- A. Only red
- B. Only blue
- C. Mixed
- D. Cannot be determined
- E. Impossible information

Question 3. A trader executes 40% of an order at 98.5, 35% at 101.2, and the rest at 99.8. What is the volume-weighted average price?

- A. 99.62
- B. 99.67
- C. 99.72
- D. 99.77
- E. 99.82

Question 4. You have a 3-liter jug, a 5-liter jug, and unlimited water. A move is fill, empty, or pour until one jug is empty or the other full. What is the minimum number of moves to measure exactly 4 liters in the 5-liter jug, starting empty?

- A. 5
- B. 6
- C. 7
- D. 8
- E. Impossible

Question 5. Each term after the first is previous term plus the number of letters in the English spelling of the previous term, ignoring spaces and hyphens. Starting at 10: 10, 13, 21, 30, ?. What is next?

- A. 35
- B. 36

- C. 38
- D. 41
- E. 44

Question 6. Count pairs (a,b) in the list below where a appears before b, a is odd, b is even, and $b-a = 5$.
List: 14, 9, 3, 8, 11, 16, 21, 26, 6, 1, 18, 23.

- A. 3
- B. 4
- C. 5
- D. 6
- E. 7

Question 7. A biased coin has $P(H)=0.6$. It is flipped until the first tail or until 4 flips have occurred, whichever comes first. What is the expected number of heads observed?

- A. 1.176
- B. 1.224
- C. 1.296
- D. 1.344
- E. 1.440

Question 8. In a tournament, each of A, B, C, D plays each other once. There are no draws. A beat exactly two players. B lost to C. D beat A. C did not have the highest number of wins. Who must have beaten D?

- A. A only
- B. B only
- C. C only
- D. A and B
- E. Cannot be determined

Question 9. A latency drops from 250 ms to 190 ms, then rises by 15% from the new level. Compared with the original, the final latency is:

- A. 8.0% lower
- B. 10.5% lower
- C. 12.6% lower
- D. 14.0% lower
- E. 15.0% lower

Question 10. A quote tape shows: U U D D U D U D D U D U. Treat U as +1 and D as -1. Starting at 0, how many times after a tick is the cumulative sum positive and immediately followed by a tick that makes it non-positive?

- A. 1
- B. 2
- C. 3
- D. 4

Optiver Cognitive Assessment - Section 4

Instructions: Choose the best answer. No calculators. Work quickly but carefully.

Question 1. A three-dial lock starts at 000. One move increases exactly one dial by 1 mod 10, or decreases exactly two different dials by 1 mod 10. What is the minimum number of moves to reach 347?

- A. 7
- B. 8
- C. 9
- D. 10
- E. 11

Question 2. From numbers 1 to 12, two distinct numbers are chosen uniformly. What is the probability their sum is prime?

- A. 13/66
- B. 5/22
- C. 3/11
- D. 19/66
- E. 7/22

Question 3. A, B, C, D each say one statement. A: B is lying. B: C is lying. C: D is lying. D: A and C have the same truth value. If exactly two statements are true, whose statement is false?

- A. A and B
- B. A and C
- C. B and C
- D. B and D
- E. C and D

Question 4. A spread is 48 bps. It tightens by 12.5%, then widens by 25% of the new spread, then tightens by 4 bps. Final spread?

- A. 47.5 bps
- B. 48.0 bps
- C. 48.5 bps
- D. 49.0 bps
- E. 50.5 bps

Question 5. Grid rule: each row's third number equals twice the first plus the second minus 1. Row1: 4,7,14. Row2: 6,5,16. Row3: 9,?,28. What is ?

- A. 8
- B. 9
- C. 10
- D. 11

E. 12

Question 6. Count the number of symbols that are immediately preceded by a digit and immediately followed by a letter: A3#b 7@C D\$4 9%x m2&N P*Q 5!r s6?t

- A. 3
- B. 4
- C. 5
- D. 6
- E. 7

Question 7. An urn has 3 black, 3 white, 2 gold balls. Draw 3 without replacement. What is the probability that exactly two colors appear and one of those colors is gold?

- A. $9/28$
- B. $5/14$
- C. $11/28$
- D. $3/7$
- E. $13/28$

Question 8. A password is a permutation of digits 1,2,3,4,5. The first digit is not 1 or 5. The number formed by the first two digits is divisible by 3. The last two digits form a number divisible by 4. The middle digit is not 1. Which digit is in the middle?

- A. 1
- B. 2
- C. 3
- D. 4
- E. 5

Question 9. A fund has 60% in asset X and 40% in Y. X rises 8%, Y falls 5%. Then the fund pays a 1% fee on the new total. Net return after fee is closest to:

- A. +1.77%
- B. +1.95%
- C. +2.20%
- D. +2.77%
- E. +3.00%

Question 10. You need exactly 11 units. Machines produce batches of 4 or 7. You may also scrap exactly 3 units once, after any production. What is the fewest operations, counting each production and the scrap as operations?

- A. 2
- B. 3
- C. 4
- D. 5

E. Impossible

Optiver Cognitive Assessment - Section 5

Instructions: Choose the best answer. No calculators. Work quickly but carefully.

Question 1. A classifier flags 20% of trades. Flagged trades are bad with probability 0.30; unflagged trades are bad with probability 0.05. A randomly chosen bad trade is inspected. What is the probability it was flagged?

- A. $\frac{3}{5}$
- B. $\frac{2}{3}$
- C. $\frac{3}{4}$
- D. $\frac{6}{7}$
- E. $\frac{5}{8}$

Question 2. In the following tokens, count those with vowels in strictly alphabetical order and consonants not repeated: BEA, ADO, FIO, UEA, ACE, EGI, APU, OIL, AET, IOU.

- A. 4
- B. 5
- C. 6
- D. 7
- E. 8

Question 3. Four servers S1-S4 processed jobs. Exactly one server crashed. The crash produces exactly the error logs specified by the matching rule, and no other error logs. If S1 crashed, S3 logged an error. If S2 crashed, S4 logged an error. If S3 crashed, S1 logged an error. If S4 crashed, S2 and S3 logged errors. Observed: exactly S2 and S3 logged errors. Which crashed?

- A. S1
- B. S2
- C. S3
- D. S4
- E. Cannot be determined

Question 4. A value is multiplied by 1.04, then by 0.97, then by 1.06. The net multiplier is closest to:

- A. 1.065
- B. 1.069
- C. 1.073
- D. 1.079
- E. 1.083

Question 5. Sequence: 5, 11, 23, 47, 95, ?. Each term doubles the previous and alternates +1, +1, +1, +1 after the first? But one visible distractor is powers of two minus one. Apply the actual rule: $\text{term}_n = 2 * \text{term}_{n-1} + 1$. What is next?

- A. 189
- B. 190

- C. 191
- D. 192
- E. 193

Question 6. A token starts at 0. A move can add 5, subtract 2, or double the current value. Values must stay between 0 and 20 inclusive. Minimum moves to reach 19?

- A. 4
- B. 5
- C. 6
- D. 7
- E. Impossible

Question 7. You draw two cards from a mini-deck containing A,A,K,K,Q,Q. You are told at least one card is A. What is the probability both are A?

- A. $1/9$
- B. $1/7$
- C. $1/6$
- D. $1/5$
- E. $2/9$

Question 8. A code uses three different letters from {A,B,C,D,E}. Rule 1: If A is used, C is not used. Rule 2: If B is used, D is used. Rule 3: E is used iff A is not used. How many valid three-letter sets exist?

- A. 1
- B. 2
- C. 3
- D. 4
- E. 5

Question 9. A trade has profit +18 on 40% of days, -12 on 35% of days, and +4 on the rest. What is expected profit per day?

- A. 3.8
- B. 4.0
- C. 4.2
- D. 4.4
- E. 4.6

Question 10. Read the sequence: 2A, 4B, 8C, 16D, 31E, 64F, 128G, 257H. Count entries whose number is a power of 2 and whose letter's alphabet position is one more than $\log_2(\text{number})$.

- A. 3
- B. 4
- C. 5
- D. 6

Optiver Cognitive Assessment - Section 6

Instructions: Choose the best answer. No calculators. Work quickly but carefully.

Question 1. A delivery route must visit A, B, C, D starting at S and ending at T. Travel times: S-A 4, S-B 6, S-C 5, S-D 7; A-B 3, A-C 8, A-D 6; B-C 4, B-D 5; C-D 2; A-T 7, B-T 5, C-T 6, D-T 4. What is the minimum route time?

- A. 17
- B. 18
- C. 19
- D. 20
- E. 21

Question 2. A fair die is rolled until a 6 appears or until three rolls are completed. You receive the maximum number rolled. What is the probability your payoff is exactly 5?

- A. $61/216$
- B. $25/72$
- C. $91/216$
- D. $1/3$
- E. $5/18$

Question 3. A, B, C, D are assigned distinct ranks 1-4, where 1 is best. A is not first. B is better than C. D is worse than A. C is not fourth. Which rank must B have?

- A. 1
- B. 2
- C. 3
- D. 4
- E. Cannot be determined

Question 4. A basket has 30 units at 42, 50 units at 38, and 20 units at 45. It sells 40 units at 43 and the rest at 39. What is total P&L?

- A. +20
- B. +40
- C. +60
- D. +80
- E. +100

Question 5. Count words where the second letter alphabetically comes after the fourth letter, and the word has exactly one repeated letter: TRADER, SIGNAL, MARKET, OPTION, SPREAD, DELTA, GAMMA.

- A. 1
- B. 2
- C. 3

D. 4

E. 5

Question 6. Matrix row rule: the fourth entry equals (first x second) - third. Row1: 3,5,4,11. Row2: 6,4,9,15. Row3: 7,?,13,22. What is ?

A. 4

B. 5

C. 6

D. 7

E. 8

Question 7. A bag has 4 green, 4 red, 4 blue chips. Draw two without replacement. What is the probability the chips have different colors and neither is blue?

A. $4/33$

B. $8/33$

C. $16/33$

D. $2/11$

E. $1/3$

Question 8. Three switches control three lamps in another room. You may inspect the room once. But each switch can be flipped at most twice, and one switch is faulty: flipping it changes nothing. Initially all lamps are off. Which strategy guarantees identifying the faulty switch and all mappings?

A. Turn one switch on, wait, turn it off; turn second on; inspect

B. Turn all switches on; inspect

C. Turn first on; second on; inspect

D. Turn first on then off; inspect

E. No strategy can guarantee this under the faulty-switch condition

Question 9. A number is increased by 25%, then the new number is decreased by $x\%$ to return to 90% of the original. What is x ?

A. 24%

B. 26%

C. 28%

D. 30%

E. 32%

Question 10. In a noisy feed, valid trades match pattern Letter-Digit-Digit-Letter, where the two digits sum to 9 and the letters are consecutive alphabetically in either order. Count valid trades: A45B B54A C63D D27C E18F F81E G90H H09G J36K K72J L44M

A. 5

B. 6

C. 7

D. 8

E. 10